Table 6: Simulation V: Empirical power and Monte Carlo standard error for $x_0 = 0.25$, under different values of δ, for $r = 2, 5, 8$.

$\delta = 0.05$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0538 (0.0032)	0.0516 (0.0031)	50	0.0000 (0.0000)	0.0480 (0.0030)	0.0498 (0.0031)	50	0.0000 (0.0000)	0.0492 (0.0031)	0.0582 (0.0033)
100	0.0000 (0.0000)	0.0550 (0.0032)	0.0464 (0.0030)	100	0.0000 (0.0000)	0.0488 (0.0030)	0.0480 (0.0030)	100	0.0000 (0.0000)	0.0568 (0.0033)	0.0494 (0.0031)
200	0.0000 (0.0000)	0.0596 (0.0033)	0.0392 (0.0027)	200	0.0000 (0.0000)	0.0552 (0.0032)	0.0510 (0.0031)	200	0.0000 (0.0000)	0.0466 (0.0030)	0.0478 (0.0030)
500	0.0000 (0.0000)	0.0536 (0.0032)	0.0382 (0.0027)	500	0.0000 (0.0000)	0.0588 (0.0033)	0.0466 (0.0030)	500	0.0000 (0.0000)	0.0584 (0.0033)	0.0496 (0.0031)
1000	0.0000 (0.0000)	0.0744 (0.0037)	0.0366 (0.0027)	1000	0.0000 (0.0000)	0.0678 (0.0036)	0.0402 (0.0028)	1000	0.0000 (0.0000)	0.0742 (0.0037)	0.0414 (0.0028)

$\delta = 0.10$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0550 (0.0032)	0.0456 (0.0030)	50	0.0000 (0.0000)	0.0522 (0.0031)	0.0476 (0.0030)	50	0.0000 (0.0000)	0.0422 (0.0028)	0.0652 (0.0035)
100	0.0000 (0.0000)	0.0564 (0.0033)	0.0438 (0.0029)	100	0.0000 (0.0000)	0.0458 (0.0030)	0.0516 (0.0031)	100	0.0000 (0.0000)	0.0492 (0.0031)	0.0548 (0.0032)
200	0.0000 (0.0000)	0.0676 (0.0036)	0.0336 (0.0025)	200	0.0000 (0.0000)	0.0542 (0.0032)	0.0476 (0.0030)	200	0.0000 (0.0000)	0.0550 (0.0032)	0.0448 (0.0029)
500	0.0000 (0.0000)	0.0668 (0.0035)	0.0346 (0.0026)	500	0.0000 (0.0000)	0.0636 (0.0035)	0.0468 (0.0030)	500	0.0000 (0.0000)	0.0606 (0.0034)	0.0490 (0.0031)
1000	0.0000 (0.0000)	0.0986 (0.0042)	0.0284 (0.0023)	1000	0.0000 (0.0000)	0.0754 (0.0037)	0.0348 (0.0026)	1000	0.0000 (0.0000)	0.0692 (0.0036)	0.0400 (0.0028)

$\delta = 0.20$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0544 (0.0032)	0.0486 (0.0030)	50	0.0000 (0.0000)	0.0492 (0.0031)	0.0498 (0.0031)	50	0.0000 (0.0000)	0.0430 (0.0029)	0.0776 (0.0038)
100	0.0000 (0.0000)	0.0608 (0.0034)	0.0418 (0.0028)	100	0.0000 (0.0000)	0.0468 (0.0030)	0.0562 (0.0033)	100	0.0000 (0.0000)	0.0448 (0.0029)	0.0576 (0.0033)
200	0.0000 (0.0000)	0.0724 (0.0037)	0.0352 (0.0026)	200	0.0000 (0.0000)	0.0510 (0.0031)	0.0500 (0.0031)	200	0.0000 (0.0000)	0.0380 (0.0027)	0.0606 (0.0034)
500	0.0000 (0.0000)	0.0876 (0.0040)	0.0244 (0.0022)	500	0.0000 (0.0000)	0.0666 (0.0035)	0.0404 (0.0028)	500	0.0000 (0.0000)	0.0450 (0.0029)	0.0582 (0.0033)
1000	0.0000 (0.0000)	0.1354 (0.0048)	0.0204 (0.0020)	1000	0.0000 (0.0000)	0.0782 (0.0038)	0.0362 (0.0026)	1000	0.0000 (0.0000)	0.0590 (0.0033)	0.0506 (0.0031)

$\delta = 0.40$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0626 (0.0034)	0.0458 (0.0030)	50	0.0000 (0.0000)	0.0396 (0.0028)	0.0718 (0.0037)	50	0.0000 (0.0000)	0.0190 (0.0019)	0.1148 (0.0045)
100	0.0000 (0.0000)	0.0748 (0.0037)	0.0436 (0.0029)	100	0.0000 (0.0000)	0.0388 (0.0027)	0.0742 (0.0037)	100	0.0000 (0.0000)	0.0276 (0.0023)	0.0868 (0.0040)
200	0.0000 (0.0000)	0.0842 (0.0039)	0.0386 (0.0027)	200	0.0000 (0.0000)	0.0450 (0.0029)	0.0664 (0.0035)	200	0.0000 (0.0000)	0.0218 (0.0021)	0.0918 (0.0041)
500	0.0000 (0.0000)	0.1056 (0.0043)	0.0278 (0.0023)	500	0.0000 (0.0000)	0.0446 (0.0029)	0.0628 (0.0034)	500	0.0000 (0.0000)	0.0198 (0.0020)	0.0992 (0.0042)
1000	0.0000 (0.0000)	0.1694 (0.0053)	0.0154 (0.0017)	1000	0.0000 (0.0000)	0.0394 (0.0028)	0.0610 (0.0034)	1000	0.0000 (0.0000)	0.0196 (0.0020)	0.1114 (0.0044)

Table 7: Simulation V: Empirical power and Monte Carlo standard error for $x_0 = 0.50$, under different values of δ, for $r = 2, 5, 8$.

$\delta = 0.05$

$r = 2$	Uniform	Central	Tail	$r = 5$	Uniform	Central	Tail	$r = 8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0498 (0.0031)	0.0516 (0.0031)	50	0.0000 (0.0000)	0.0552 (0.0032)	0.0468 (0.0030)	50	0.0000 (0.0000)	0.0482 (0.0030)	0.0544 (0.0032)
100	0.0000 (0.0000)	0.0482 (0.0030)	0.0540 (0.0032)	100	0.0000 (0.0000)	0.0540 (0.0032)	0.0492 (0.0031)	100	0.0000 (0.0000)	0.0552 (0.0032)	0.0438 (0.0029)
200	0.0000 (0.0000)	0.0464 (0.0030)	0.0532 (0.0032)	200	0.0000 (0.0000)	0.0484 (0.0030)	0.0592 (0.0033)	200	0.0000 (0.0000)	0.0562 (0.0033)	0.0504 (0.0031)
500	0.0000 (0.0000)	0.0448 (0.0029)	0.0506 (0.0031)	500	0.0000 (0.0000)	0.0502 (0.0031)	0.0522 (0.0031)	500	0.0000 (0.0000)	0.0512 (0.0031)	0.0520 (0.0031)
1000	0.0000 (0.0000)	0.0450 (0.0029)	0.0520 (0.0031)	1000	0.0000 (0.0000)	0.0532 (0.0032)	0.0466 (0.0030)	1000	0.0000 (0.0000)	0.0532 (0.0032)	0.0454 (0.0029)

$\delta = 0.10$

$r = 2$	Uniform	Central	Tail	$r = 5$	Uniform	Central	Tail	$r = 8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0520 (0.0031)	0.0492 (0.0031)	50	0.0000 (0.0000)	0.0586 (0.0033)	0.0386 (0.0027)	50	0.0000 (0.0000)	0.0508 (0.0031)	0.0514 (0.0031)
100	0.0000 (0.0000)	0.0540 (0.0032)	0.0546 (0.0032)	100	0.0000 (0.0000)	0.0536 (0.0032)	0.0474 (0.0030)	100	0.0000 (0.0000)	0.0540 (0.0032)	0.0516 (0.0031)
200	0.0000 (0.0000)	0.0552 (0.0032)	0.0510 (0.0031)	200	0.0000 (0.0000)	0.0514 (0.0031)	0.0564 (0.0033)	200	0.0000 (0.0000)	0.0544 (0.0032)	0.0536 (0.0032)
500	0.0000 (0.0000)	0.0482 (0.0030)	0.0536 (0.0032)	500	0.0000 (0.0000)	0.0540 (0.0032)	0.0578 (0.0033)	500	0.0000 (0.0000)	0.0598 (0.0034)	0.0572 (0.0033)
1000	0.0000 (0.0000)	0.0570 (0.0033)	0.0624 (0.0034)	1000	0.0000 (0.0000)	0.0638 (0.0035)	0.0532 (0.0032)	1000	0.0000 (0.0000)	0.0690 (0.0036)	0.0598 (0.0034)

$\delta = 0.20$

$r = 2$	Uniform	Central	Tail	$r = 5$	Uniform	Central	Tail	$r = 8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0680 (0.0036)	0.0536 (0.0032)	50	0.0000 (0.0000)	0.0680 (0.0036)	0.0436 (0.0029)	50	0.0000 (0.0000)	0.0558 (0.0032)	0.0520 (0.0031)
100	0.0000 (0.0000)	0.0698 (0.0036)	0.0548 (0.0032)	100	0.0000 (0.0000)	0.0584 (0.0033)	0.0554 (0.0032)	100	0.0000 (0.0000)	0.0602 (0.0034)	0.0456 (0.0030)
200	0.0000 (0.0000)	0.0618 (0.0034)	0.0552 (0.0032)	200	0.0000 (0.0000)	0.0622 (0.0034)	0.0668 (0.0035)	200	0.0000 (0.0000)	0.0612 (0.0034)	0.0600 (0.0034)
500	0.0000 (0.0000)	0.0546 (0.0032)	0.0570 (0.0033)	500	0.0000 (0.0000)	0.0690 (0.0036)	0.0714 (0.0036)	500	0.0000 (0.0000)	0.0754 (0.0037)	0.0642 (0.0035)
1000	0.0000 (0.0000)	0.0732 (0.0037)	0.0640 (0.0035)	1000	0.0000 (0.0000)	0.0724 (0.0037)	0.0640 (0.0035)	1000	0.0000 (0.0000)	0.0876 (0.0040)	0.0682 (0.0036)

$\delta = 0.40$

$r = 2$	Uniform	Central	Tail	$r = 5$	Uniform	Central	Tail	$r = 8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0926 (0.0041)	0.0380 (0.0027)	50	0.0000 (0.0000)	0.0706 (0.0036)	0.0232 (0.0021)	50	0.0000 (0.0000)	0.0478 (0.0030)	0.0316 (0.0025)
100	0.0000 (0.0000)	0.0938 (0.0041)	0.0558 (0.0032)	100	0.0000 (0.0000)	0.0770 (0.0038)	0.0392 (0.0027)	100	0.0000 (0.0000)	0.0650 (0.0035)	0.0250 (0.0022)
200	0.0000 (0.0000)	0.0962 (0.0042)	0.0674 (0.0035)	200	0.0000 (0.0000)	0.0824 (0.0039)	0.0606 (0.0034)	200	0.0000 (0.0000)	0.0686 (0.0036)	0.0394 (0.0028)
500	0.0000 (0.0000)	0.0950 (0.0041)	0.0744 (0.0037)	500	0.0000 (0.0000)	0.0966 (0.0042)	0.0690 (0.0036)	500	0.0000 (0.0000)	0.0794 (0.0038)	0.0626 (0.0034)
1000	0.0000 (0.0000)	0.1068 (0.0044)	0.0914 (0.0041)	1000	0.0000 (0.0000)	0.0928 (0.0041)	0.0666 (0.0035)	1000	0.0000 (0.0000)	0.0796 (0.0038)	0.0594 (0.0033)

Table 8: Simulation V: Empirical power and Monte Carlo standard error for $x_0 = 0.75$, under different values of δ, for $r = 2, 5, 8$.

$\delta = 0.05$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0444 (0.0029)	0.0578 (0.0033)	50	0.0000 (0.0000)	0.0472 (0.0030)	0.0394 (0.0028)	50	0.0000 (0.0000)	0.0394 (0.0028)	0.0596 (0.0033)
100	0.0000 (0.0000)	0.0416 (0.0028)	0.0706 (0.0036)	100	0.0000 (0.0000)	0.0318 (0.0025)	0.0718 (0.0037)	100	0.0000 (0.0000)	0.0418 (0.0028)	0.0576 (0.0033)
200	0.0000 (0.0000)	0.0284 (0.0023)	0.0846 (0.0039)	200	0.0000 (0.0000)	0.0280 (0.0023)	0.0926 (0.0041)	200	0.0000 (0.0000)	0.0302 (0.0024)	0.0826 (0.0039)
500	0.0000 (0.0000)	0.0136 (0.0016)	0.1108 (0.0044)	500	0.0000 (0.0000)	0.0194 (0.0020)	0.1208 (0.0046)	500	0.0000 (0.0000)	0.0200 (0.0020)	0.1306 (0.0048)
1000	0.0000 (0.0000)	0.0104 (0.0014)	0.1650 (0.0052)	1000	0.0000 (0.0000)	0.0122 (0.0016)	0.1470 (0.0050)	1000	0.0000 (0.0000)	0.0126 (0.0016)	0.1676 (0.0053)

$\delta = 0.10$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0326 (0.0025)	0.0774 (0.0038)	50	0.0000 (0.0000)	0.0320 (0.0025)	0.0456 (0.0030)	50	0.0000 (0.0000)	0.0322 (0.0025)	0.0446 (0.0029)
100	0.0000 (0.0000)	0.0124 (0.0016)	0.1232 (0.0046)	100	0.0000 (0.0000)	0.0162 (0.0018)	0.1084 (0.0044)	100	0.0000 (0.0000)	0.0216 (0.0021)	0.0926 (0.0041)
200	0.0000 (0.0000)	0.0086 (0.0013)	0.1848 (0.0055)	200	0.0000 (0.0000)	0.0082 (0.0013)	0.1984 (0.0056)	200	0.0000 (0.0000)	0.0080 (0.0013)	0.1768 (0.0054)
500	0.0000 (0.0000)	0.0026 (0.0007)	0.3300 (0.0066)	500	0.0000 (0.0000)	0.0012 (0.0005)	0.3452 (0.0067)	500	0.0000 (0.0000)	0.0014 (0.0005)	0.3708 (0.0068)
1000	0.0000 (0.0000)	0.0000 (0.0000)	0.5338 (0.0071)	1000	0.0000 (0.0000)	0.0000 (0.0000)	0.5426 (0.0070)	1000	0.0000 (0.0000)	0.0000 (0.0000)	0.5688 (0.0070)

$\delta = 0.20$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0190 (0.0019)	0.1614 (0.0052)	50	0.0000 (0.0000)	0.0214 (0.0020)	0.0528 (0.0032)	50	0.0000 (0.0000)	0.0154 (0.0017)	0.0400 (0.0028)
100	0.0000 (0.0000)	0.0022 (0.0007)	0.3492 (0.0067)	100	0.0000 (0.0000)	0.0006 (0.0003)	0.2960 (0.0065)	100	0.0000 (0.0000)	0.0010 (0.0004)	0.1632 (0.0052)
200	0.0000 (0.0000)	0.0002 (0.0002)	0.6356 (0.0068)	200	0.0000 (0.0000)	0.0002 (0.0002)	0.6506 (0.0067)	200	0.0000 (0.0000)	0.0000 (0.0000)	0.5752 (0.0070)
500	0.0000 (0.0000)	0.0000 (0.0000)	0.9442 (0.0032)	500	0.0000 (0.0000)	0.0000 (0.0000)	0.9672 (0.0025)	500	0.0000 (0.0000)	0.0000 (0.0000)	0.9654 (0.0026)
1000	0.0000 (0.0000)	0.0000 (0.0000)	0.9988 (0.0005)	1000	0.0000 (0.0000)	0.0000 (0.0000)	0.9996 (0.0003)	1000	0.0000 (0.0000)	0.0000 (0.0000)	0.9994 (0.0003)

$\delta = 0.40$

$r=2$	Uniform	Central	Tail	$r=5$	Uniform	Central	Tail	$r=8$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.2804 (0.0064)	0.0112 (0.0015)	50	0.0000 (0.0000)	0.2150 (0.0058)	0.0152 (0.0017)	50	0.0000 (0.0000)	0.1784 (0.0054)	0.0244 (0.0022)
100	0.0000 (0.0000)	0.4254 (0.0070)	0.0056 (0.0011)	100	0.0000 (0.0000)	0.2824 (0.0064)	0.0120 (0.0015)	100	0.0000 (0.0000)	0.2526 (0.0061)	0.0144 (0.0017)
200	0.0000 (0.0000)	0.6026 (0.0069)	0.0034 (0.0008)	200	0.0000 (0.0000)	0.4464 (0.0070)	0.0058 (0.0011)	200	0.0000 (0.0000)	0.3366 (0.0067)	0.0050 (0.0010)
500	0.0000 (0.0000)	0.8660 (0.0048)	0.0000 (0.0000)	500	0.0000 (0.0000)	0.7388 (0.0062)	0.0016 (0.0006)	500	0.0000 (0.0000)	0.6158 (0.0069)	0.0028 (0.0007)
1000	0.0000 (0.0000)	0.9792 (0.0020)	0.0000 (0.0000)	1000	0.0000 (0.0000)	0.9224 (0.0038)	0.0000 (0.0000)	1000	0.0000 (0.0000)	0.8362 (0.0052)	0.0000 (0.0000)